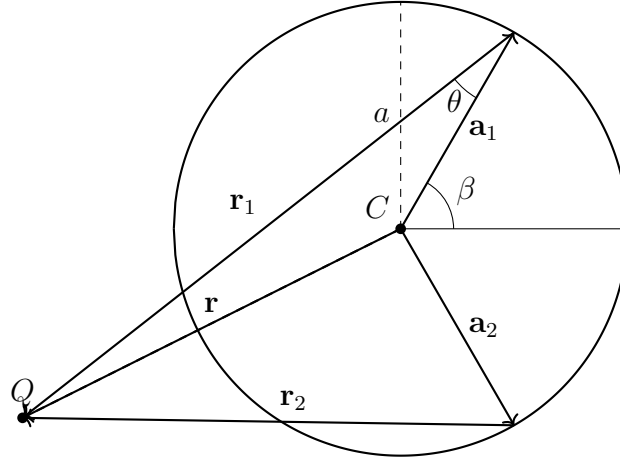




A força $d\mathbf{F}_1$ é então:

$$d\mathbf{F}_1 = k \frac{Q dq}{r^3} \mathbf{r}_1$$

Escrevendo $\mathbf{r}_1$ em termos de $\mathbf{i}$ e $\mathbf{j}$:

$$\mathbf{r}_1 = \mathbf{r} - \mathbf{a}_1$$

$$\mathbf{r}_1 = -a \cos \beta \mathbf{i} - a \sin \beta \mathbf{j} + r \mathbf{k}$$

Definindo uma distribuição de carga uniforme λ na parte superior:

$$\lambda = \frac{dq}{d\beta a} \Rightarrow dq = \lambda d\beta a$$

Ou seja:

$$d\mathbf{F}_1 = k \frac{Q \lambda d\beta a}{r^3} (-a \cos \beta \mathbf{i} - a \sin \beta \mathbf{j} + r \mathbf{k})$$

$$\mathbf{F}_1 = k \frac{Q \lambda a}{r^3} \int_0^\pi d\beta (-a \cos \beta \mathbf{i} - a \sin \beta \mathbf{j} + r \mathbf{k})$$

$$\mathbf{F}_1 = k \frac{Q \lambda a}{r^3} \left[\int_0^\pi -a \cos \beta d\beta + \int_0^\pi -a \sin \beta d\beta \mathbf{j} + \int_0^\pi r d\beta \mathbf{k} \right]$$

$$\mathbf{F}_1 = k \frac{Q \lambda a}{r^3} [-a \sin \beta \Big|_0^\pi - (-\cos \beta \Big|_0^\pi) + \beta \Big|_0^\pi]$$